

Airbag ECU Portfolio — HS Harz — Functional Safety

AIRBAG ELECTRONIC CONTROL UNIT

Measurement and Processing of Sensor Signals to Control Airbag Operation

VERIFICATION AND VALIDATION PLAN (V&V PLAN)

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1 Document Control and Administration

1.1 Meetings within the Project

The following project meetings were held during the V&V planning phase:

Table 1: Meetings within the Project

Topic	Date / Time	Participants	Minutes / Decisions
V&V Plan Kick-off and Scope Review	2026-05-15, 10:00	Testing Engineer V&V, Functional Safety Manager, Software Architect, Quality Manager	Confirmed FSR scope (FSR-01 to FSR-22). Reviewed SADS-AIRBAG-ECU-001 Rev 2.0 test cases TC-01 to TC-18. Assigned verification and validation task ownership.
V&V Plan Review and Gap Assessment	2026-06-10, 14:00	Testing Engineer V&V, Safety Requirement Engineers, Quality Manager, Project Manager	Reviewed compliance gaps GAP-01 to GAP-07 from SADS Rev 2.0. Confirmed impact on testability. Test cases updated to reflect gap status. Plan approved for DRAFT release.

1.2 Unresolved Topics

Table 2: Unresolved Topics

No.	Topic / Question	Responsible	Target Date	Status	Notes
1	FSR-15 CAN timeout: FSRS specifies 50 ms; SADS implements 2000 ms. Discrepancy must be resolved before final release.	Software Safety Engineer	2026-06-30	OPEN	See Section 4.4 (GAP-15 notation in RTM). Acceptance rationale required or firmware update needed.
2	GAP-01 to GAP-07 (SADS Rev 2.0): Seven firmware compliance gaps. Verification of FSR-11, FSR-12, FSR-13, FSR-14, FSR-21, FSR-22, POST CAN check remain blocked pending gap resolution.	Software Safety Engineer / V&V Lead	2026-07-15	OPEN	Test cases referencing gaps are marked *GAP-xx and cannot achieve PASS status until firmware is updated.

1.3 Glossary

Table 3 lists all abbreviations and terms used in this document and the wider Airbag ECU project portfolio.

Table 3: Glossary

Term / Abbreviation	Definition
ASIL	Automotive Safety Integrity Level — risk classification per ISO 26262 (QM, A, B, C, D)
ACU	Airbag Control Unit — synonymous with Airbag ECU

BB	Black Box — test method that exercises inputs/outputs without knowledge of internal logic
CAN	Controller Area Network — serial communication bus for vehicle ECUs
CRC	Cyclic Redundancy Check — error-detection mechanism for data integrity verification
DTC	Diagnostic Trouble Code
DWI	Diagnostic Warning Indicator — airbag fault lamp in the instrument cluster
ECU	Electronic Control Unit
EMC	Electromagnetic Compatibility
FSR	Functional Safety Requirement — from LCP04-AIRBAG-ECU-001 FSRs
FSRS	Functional Safety Requirements Specification
GAP	Compliance gap identified in SADS-AIRBAG-ECU-001 Rev 2.0 between specification and implementation
HARA	Hazard and Risk Analysis
HW	Hardware
IEC 61508	Functional Safety of E/E/PE Safety-related Systems (2010 edition)
ISO 26262	Road Vehicles — Functional Safety standard (2018 edition)
MC/DC	Modified Condition / Decision Coverage — mandatory for ASIL D software per ISO 26262 Part 6
MCU	Microcontroller Unit
MOD	Software module identifier from SADS-AIRBAG-ECU-001 (MOD-01 through MOD-06)
NCR	Non-Conformance Report
OCS	Occupant Classification System
POST	Power-On Self-Test
RTM	Requirements Traceability Matrix
SADS	Software Architecture and Design Specification — SADS-AIRBAG-ECU-001
SCD	Safety Concept Document — LCP05-AIRBAG-ECU-001
SG	Safety Goal — high-level safety requirement derived from HARA
SIL	Safety Integrity Level per IEC 61508 (1–4)
SPFM	Single-Point Fault Metric
SS	Safe State — defined system condition upon fault detection (SS-01 through SS-05)
SW	Software
TC	Test Case
TC-VAL	Validation test case (system-level black-box)
TC-VER	Verification test case (unit or integration level)
V&V	Verification and Validation
WB	White Box — test method that exercises internal code paths and conditions

WDT	Watchdog Timer
2oo2	Two-out-of-Two voting architecture — both channels must agree for deployment

1.4 Organisational Structure and Role Separation

In accordance with ISO 26262 Part 8 and IEC 61508-1 Clause 6, verification and validation activities shall be conducted by personnel independent of the development team. The following virtual organisational structure enforces this separation:

Table 4: Organisational Roles and Independence

Role	Responsibility	Independence Level
Functional Safety Manager (Kruthika S. Gowda)	Overall V&V programme governance; sign-off authority on all test reports; ASIL D compliance oversight.	Independent of SW/HW development
Testing Engineer V&V (Shivbaraje Bhosale)	V&V plan management; test execution lead; traceability matrix maintenance; test environment configuration control.	Independent of SW/HW development
Quality Manager (Thirthashri G. Ganesh)	Execution oversight; recording actual results; issuing pass/fail verdicts; NCR management.	Independent of SW development
Software Architect (Mariana Chavez Avila)	Unit code authoring; module implementation; embedded algorithm development.	Development role — not permitted to execute own test cases
System Architect / HW Engineer (Om Naik / Atharva Avsare)	Breadboard wiring verification; ADC signal calibration; hardware fault injection setup.	Independent of SW development
Safety Requirement Engineers (Fateme Hosseinzadeh, Mobina Ketabi)	Safety analysis support; FSRS clarification; traceability review.	Independent of SW development
Project Supervisor	Final approval authority; regulatory compliance review.	Senior independent oversight

2 Scope and Objectives

2.1 Item Boundary

This V&V Plan applies to the Airbag ECU laboratory implementation as defined in LCP01-AIRBAG-ECU-001 and SADS-AIRBAG-ECU-001 Rev 2.0. The item under verification comprises:

- Two Arduino Due (SAM3X8E, 84 MHz ARM Cortex-M3) microcontroller boards operating in a 2oo2 dual-channel redundant architecture
- Three potentiometer-simulated analog crash sensors on a breadboard: POTI-1 (accelerometer, pin A0), POTI-2 (gyroscope, pin A1), POTI-3 (pressure sensor, pin A2) — all with 12-bit ADC resolution (0–4095 counts, 0–3.3 V)
- One digital baby-seat switch input (active-LOW, 10 kΩ pull-up to 3.3 V, digital pin D11)
- CAN bus inter-ECU communication (500 kbit/s, Waveshare SN65HVD230 transceivers; ECU1 TX ID 0x100 / RX 0x200; ECU2 TX ID 0x200 / RX 0x100)
- CAN bus output to OMS laboratory PC via USB-CAN-A interface (USBCANv2 at 500 kbit/s)
- Hardware AND-gate (DM74LS08N) forming the physical 2oo2 deployment enforcement layer
- Embedded firmware implementing MOD-01 (POST), MOD-02 (Sensor Acquisition), MOD-03 (Crash Detection), MOD-04 (Deployment Control), MOD-05 (CAN Communication), MOD-06 (Diagnostic Monitor)

The following are outside the item boundary and not covered by this V&V Plan:

- Physical pyrotechnic inflator, airbag cushion, squib circuits, and vehicle body structure
- OMS laboratory PC software beyond CAN frame logging
- Occupant Classification System (OCS) hardware (represented by manual digital switch)

2.2 Verification Objectives (IEC 61508 Phases 6 and 9)

Verification confirms that the Arduino Due implementation correctly realises the requirements stated in LCP04-AIRBAG-ECU-001 (FSRS v1.00). The following shall be verified:

- All 22 functional safety requirements (FSR-01 through FSR-22) are correctly implemented in hardware and software
- The 2oo2 dual-channel architecture enforces AND-gate deployment authorisation as specified in FSR-03 and FSR-08
- The crash detection algorithm executes within the 1 ms deterministic cycle time (FSR-02)
- POST completes within 800 ms and verifies CPU, RAM, ADC, clock, CAN, baby-seat, power supply, and DWI (FSR-17)
- Continuous background diagnostics execute at intervals not exceeding 100 ms (FSR-18)
- Memory integrity protection mechanisms (CRC32, stack guard, WDT) are operational (FSR-22)
- Baby-seat suppress-on-fault strategy defaults to passenger airbag suppression on signal loss (FSR-10)

2.3 Validation Objectives (IEC 61508 Phases 7 and 8)

Validation confirms that the integrated Airbag ECU system correctly mitigates the real-world hazards identified in LCP03-AIRBAG-ECU-001 (HARA v2.00). The following shall be validated against safety goals:

- SG-01 / H1-01, H1-02: The system does NOT deploy airbags in the absence of a valid crash event
- SG-02 / H2-01, H2-02: The system correctly deploys airbags within the required timing window during simulated crash events
- SG-03 / H1-04: The passenger airbag is suppressed when the baby-seat digital input is active (LOW) or when the input line is faulted (suppress-on-fault)
- SG-05 / H1-05: EMC disturbances and single-channel sensor spikes do not produce a deployment command
- SG-06 / H4-01: The DWI activates within 800 ms of any safety-critical fault detection

- SG-07 / H4-02: CAN crash notification frames are transmitted within 15 ms of deployment authorisation (note: FSR-16); CAN timeout detection per FSR-15 is subject to open deviation — see Section 4.4

2.4 Known Compliance Gaps Affecting V&V Scope

IMPORTANT: SADS-AIRBAG-ECU-001 Rev 2.0 Section 7 identifies seven firmware compliance gaps (GAP-01 to GAP-07) between the current Arduino Due implementation and the FSR requirements. These gaps directly affect the achievability of PASS verdicts for the associated test cases. All gap-affected test cases are marked *GAP-xx in the RTM (Section 8) and in test case descriptions below. Gap closure is required before system-level release.

Table 5: Firmware Compliance Gaps Affecting V&V

Gap ID	FSR	Description	Risk	Required Action
GAP-01	FSR-11	Baby seat debounce not implemented — D11 read directly without 10 ms stability filter	Noisy switch could cause spurious state change	Implement debounce timer in MOD-02 (code shown in SADS Section 5.2)
GAP-02	FSR-12	Baby seat line continuity monitoring not implemented — open-circuit stuck-HIGH not detected	Open circuit mimics adult occupancy; passenger airbag not suppressed	Add periodic D11 continuity test in MOD-06
GAP-03	FSR-13	CRC not included in CAN frame — bytes 6–7 carry status and ECU_ID, not a 16-bit CRC	Corrupted frames accepted without error detection	Compute 16-bit CRC over bytes 0–5, store in bytes 6–7
GAP-04	FSR-14	Parallel CAN bitstream readback not implemented	CAN transmission corruption undetected	Implement readback channel or CAN loopback verification
GAP-05	FSR-17	CAN communication not verified during POST — Can0.begin() success not checked	Silent CAN initialisation failure allows system to proceed	Check Can0.begin() return value; set faultCOMM if it fails
GAP-06	FSR-22	Hardware Watchdog Timer not enabled in firmware	Software lockup does not trigger MCU reset — system hangs silently	Enable WDT in setup(); feed at end of every loop() iteration
GAP-07	FSR-21	Fault logging to DTC/non-volatile memory not implemented — faults shown on Serial only	Fault history lost on power cycle; cannot be retrieved by service tools	Implement DTC logging to EEPROM or flash

3 Requirements Traceability Strategy

3.1 Traceability Chain

A complete bidirectional traceability chain shall be maintained from hazard identification through to test execution:

HARA (LCP03) → Safety Goals (SG) → FSR Requirements (FSR, LCP04) → Safety Concept (SCD, LCP05) → Source Code Modules (SADS) → Test Cases (TC)

- Every ASIL D / SIL 3 safety requirement shall have at least one corresponding test case
- No test case shall exist without a traceable safety requirement
- Orphaned test cases or uncovered requirements shall be treated as non-conformances
- Gap-affected requirements (GAP-01 to GAP-07) shall be tracked with *GAP notation; they cannot be PASS until firmware is updated

3.2 Traceability Matrix Format

The master Requirements Traceability Matrix (RTM) in Section 8 uses the following column structure:

Table 6: RTM Column Definitions

RTM Column	Content / Source
Hazard ID	Source hazardous event from LCP03 HARA (e.g., H1-01, H2-02)
Safety Goal	Governing safety goal from HARA Section 10 (e.g., SG-01)
ASIL / SIL	Required integrity level for the requirement
FSR ID	Functional safety requirement from LCP04 FSRs (e.g., FSR-03)
FSR Short Title	Abbreviated requirement title for readability
SC Mechanism	Corresponding safety mechanism from LCP05 SCD
Code Module(s)	Arduino Due firmware module(s) from SADS-AIRBAG-ECU-001 Rev 2.0
Test Case ID(s)	One or more V&V test case identifiers (TC-VER-xxx or TC-VAL-xxx); *GAP-xx denotes gap dependency
Coverage Type	Unit / Integration / Black-box / Fault Injection / Timing / WB MC/DC
Status	NOT TESTED / PASS / FAIL / BLOCKED — must be NOT TESTED at plan issue

3.3 Coverage Completeness Requirement

Prior to system release, the RTM shall demonstrate:

- 100% of ASIL D / SIL 3 requirements (FSR-01 to FSR-22) covered by at least one PASS test case
- 100% of Safety Goals (SG-01 to SG-07) validated by at least one black-box system validation case
- All identified fault injection scenarios explicitly tested
- No FAIL or BLOCKED status remaining at final release
- All compliance gaps (GAP-01 to GAP-07) resolved and re-tested before status can be PASS

3.4 Code Coverage Goals

In accordance with ISO 26262 Part 6 Table 12 (ASIL D), the following code coverage targets are mandatory for all safety-relevant software modules:

Table 7: Code Coverage Targets

Coverage Metric	ASIL D Target	Applicable Modules
Statement Coverage	100%	All safety-relevant modules: MOD-01 (POST), MOD-02 (Sensor), MOD-03 (Crash Detect), MOD-04 (Deployment), MOD-05 (CAN TX), MOD-06 (Diagnostics)
Branch Coverage	100%	Crash threshold evaluation, baby-seat suppression logic, safe-state transitions, POST pass/fail paths
MC/DC Coverage	100%	Deployment authorisation 6-condition AND-gate (FSR-03); suppress-on-fault logic (FSR-10); cross-channel agreement (FSR-08)
Path Coverage	Best effort	Complex conditional paths in dual-channel cross-comparison routine

NOTE: Code coverage shall be measured using a code instrumentation approach compatible with the Arduino Due toolchain. Coverage reports shall be generated and archived as evidence artefacts for each software release.

4 Verification Plan (IEC 61508 Phases 6 and 9)

Verification activities confirm that the Arduino Due implementation correctly realises the FSRs requirements. Three levels of verification are applied: Module/Unit Testing, Integration Testing, and Timing Verification. All test cases align with and extend TC-01 through TC-18 defined in SADS-AIRBAG-ECU-001 Rev 2.0 Section 8.

4.1 Test Environment Setup

All verification tests shall be conducted in the following configured environment (per SADS Section 8.1):

- Connect both Arduino Due boards per HADS-AIRBAG-ECU-001 Rev 2.1 wiring tables; verify DM74LS08N AND gate connections at D8 output pins
- Connect OMS PC via USB-CAN-A; open USBCANv2 at 500 kbit/s, monitoring CAN IDs 0x100 and 0x200
- Open Serial Monitor (115200 baud) on both ECUs for fault flag and decision variable visibility
- Potentiometers POTI-1, POTI-2, POTI-3 set to full-minimum position (ADC ~0, below all thresholds) as default baseline
- Baby-seat switch open — D11 HIGH via pull-up to 3.3 V (adult mode) as default baseline
- Power supply: 12 V nominal; verify voltage within 9–16 V operating window before each session

Table 8: ADC Threshold Reference (12-bit, 0–4095)

Sensor	Pin	ADC Range	Engineering Range	Deploy Threshold	ADC Threshold Count
Accelerometer (POTI-1)	A0	0–4095	0–100 g	> 30.0 g	raw > 1228 (29.97% of 4095)
Gyroscope (POTI-2)	A1	0–4095	0–360 °/s	> 120.0 °/s	raw > 1365 (33.33% of 4095)
Pressure (POTI-3)	A2	0–4095	0–500 kPa	> 200.0 kPa	raw > 1638 (40.00% of 4095)
Baby-Seat Switch	D11	HIGH / LOW	TRUE / FALSE	LOW = baby seat	— (digital)

4.2 Unit / Module Test Cases

4.2.1 TC-VER-001 — ADC Raw Value to Physical Unit Scaling Verification

Table 9: TC-VER-001

Test ID	TC-VER-001
Title	ADC Raw Value to Physical Unit Scaling Verification
FSRS Reference	FSR-01, FSR-02
ASIL / SIL	ASIL D / SIL 3
Module Under Test	MOD-02 (Sensor Acquisition) — POTI-1, POTI-2, POTI-3 12-bit ADC conversion and scaling functions
Method	Unit test with stubbed ADC reads; inject raw 12-bit ADC values (0–4095 range). Verify linear scaling to engineering units.
Input Conditions	POTI-1: ADC=0 → 0 g; ADC=2048 → ~50 g; ADC=4095 → 100 g POTI-2: ADC=0 → 0 °/s; ADC=2048 → ~180 °/s; ADC=4095 → 360 °/s POTI-3: ADC=0 → 0 kPa; ADC=2048 → ~250 kPa; ADC=4095

	→ 500 kPa Threshold boundary: POTI-1 ADC=1228 (30 g), ADC=1229 (30.024 g)
Scaling Formula	$\text{accel} = (\text{rawA0} / 4095.0) \times 100.0$; $\text{gyro} = (\text{rawA1} / 4095.0) \times 360.0$; $\text{press} = (\text{rawA2} / 4095.0) \times 500.0$
Expected Output	Scaled values match linear formula within $\pm 0.5\%$ tolerance; no out-of-range values accepted; ADC values > 4095 raise faultADC flag
Pass Criterion	All nine boundary data points return correct physical values within tolerance; faultADC set for invalid ADC input; no spurious deployment flag generated
Coverage Target	100% Statement; 100% Branch (valid / invalid ADC path)
Notes	The Arduino Due uses a 12-bit ADC (analogReadResolution(12) set in setup()). The earlier revision of this plan incorrectly used 10-bit values (0–1023). All test inputs must use 12-bit range (0–4095).
Status	NOT TESTED

4.2.2 TC-VER-002 — Deployment Threshold Decision Logic (MC/DC)

Table 10: TC-VER-002

Test ID	TC-VER-002
Title	Deployment Threshold Decision Logic — Unit Verification with MC/DC
FSRS Reference	FSR-01, FSR-02, FSR-03, FSR-04
ASIL / SIL	ASIL D / SIL 3
Module Under Test	MOD-03 (Crash Detection) + MOD-04 (Deployment Control) — threshold evaluation and 6-condition voteDeploy logic
Method	Unit test with all six FSR-03 conditions exercised independently per MC/DC matrix. Each condition independently flipped to verify it independently affects the voteDeploy output. Covers SADS Section 4.3.2 and 4.4.1.
Input Conditions	Vary each of the six conditions (sensor threshold exceeded, localCrash, remoteCrash, !systemFault, !remoteFault, !babySeat) independently. Exercise all boundary combinations that determine voteDeploy = TRUE vs FALSE.
Expected Output	voteDeploy = TRUE only when ALL six conditions are simultaneously satisfied; any single condition FALSE → voteDeploy = FALSE; localCrash follows OR logic of three sensor thresholds
Pass Criterion	MC/DC matrix shows every condition independently affects deployment decision; zero false positives; zero false negatives; algorithm cycle time ≤ 1 ms measured via micros() timestamp (FSR-02)
Coverage Target	100% MC/DC mandatory for ASIL D (ISO 26262 Part 6)
Status	NOT TESTED

4.2.3 TC-VER-003 — Memory Integrity and Watchdog Supervision

Table 11: TC-VER-003

Test ID	TC-VER-003
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Title	Memory Integrity and Watchdog Supervision — Unit Verification
FSRS Reference	FSR-22
ASIL / SIL	ASIL D / SIL 3
Module Under Test	MOD-01 (POST) + MOD-06 (Diagnostics) — CRC32 RAM verification, stack boundary monitor, WDT kick routine
Method	(a) Normal RAM: CRC pass — verify no action taken. (b) Inject deliberate RAM corruption at non-critical test address; verify CRC mismatch triggers SS-05 and DWI. (c) Inhibit watchdog kick for 12 ms (> 10 ms timeout); verify MCU resets.
Input Conditions	(a) Clean system state. (b) Manually corrupt test RAM variable. (c) Disable WDT kick call in loop() via conditional compilation flag.
Expected Output	(a) CRC check PASS, no action. (b) CRC mismatch: safe state SS-05 entered, DWI HIGH, deployment inhibited, DTC logged. (c) WDT fires within 10 ms, MCU resets.
Pass Criterion	All three scenarios produce expected safe-state behaviour within required timing; DTC logged for (b) and (c)
Coverage Target	100% Branch on all CRC pass/fail and WDT paths
Notes	*GAP-06: WDT not currently enabled in firmware. Sub-test (c) is BLOCKED until GAP-06 is resolved. Sub-tests (a) and (b) can proceed independently.
Status	NOT TESTED (partially BLOCKED pending GAP-06)

4.2.4 TC-VER-004 — CAN Frame Construction and CRC Calculation

Table 12: TC-VER-004

Test ID	TC-VER-004
Title	CAN Frame Construction, Bit Assignment, and CRC Calculation
FSRS Reference	FSR-13, FSR-14, FSR-16, FSR-16A
ASIL / SIL	ASIL B / SIL 2 (ASIL D end-to-end per FSR-16A)
Module Under Test	MOD-05 (CAN Communication) — frame packing, CRC calculation (bytes 6–7), parallel readback comparison
Method	Construct known deployment state vector; call frame packing function; verify each bit field at correct byte/bit position per Table 13 (CAN frame definition); independently calculate 16-bit CRC over bytes 0–5 and compare; inject 1-bit error and verify rejection.
Input Conditions	(a) Driver deploy=1, pax deploy=0, pax suppress=1, crash=1, Ch-A fault=0, Ch-B fault=0, DWI=0, safe-state=0 — expected Byte 0 = 0x0D. (b) Same frame with 1-bit corruption in byte 3.
Expected Output	(a) Byte 0 = 0x0D; CRC field in bytes 6–7 matches independently calculated value; parallel readback passes. (b) CRC mismatch → frame rejected; CAN fault DTC raised; parallel readback discrepancy detected.
Pass Criterion	Bit fields correctly placed in all 8 bytes; CRC matches in (a); frame correctly rejected in (b); parallel readback discrepancy detected in (b)
Coverage Target	100% Statement and Branch in CAN packing and CRC routines

Notes	*GAP-03: CRC not currently implemented — bytes 6–7 carry status and ECU_ID in current firmware. *GAP-04: Parallel readback not implemented. Both sub-tests (a) CRC and (b) rejection are BLOCKED until GAP-03 and GAP-04 are resolved. Byte field assignments are per LCP04 FSRs Section 11.4.
Status	NOT TESTED (BLOCKED pending GAP-03 and GAP-04)

4.3 Integration Test Cases

4.3.1 TC-VER-005 — Analog Sensor ADC Reading Hardware Integration

Table 13: TC-VER-005

Test ID	TC-VER-005
Title	POTI Sensor Raw ADC Reading — Hardware/Software Integration
FSRS Reference	FSR-01, FSR-07
ASIL / SIL	ASIL D / SIL 3
Method	Set POTI-1 to 0 V, 1.65 V (midpoint ± 0.05 V), 3.3 V using calibrated bench power supply. Read ADC value via Serial debug output on both Channel A and Channel B Arduino Due boards independently. Repeat for POTI-2 and POTI-3.
Input Conditions	Three POTI positions: minimum (0 V $\rightarrow$ ADC ~ 0), midpoint (1.65 V $\rightarrow$ ADC ~ 2048), maximum (3.3 V $\rightarrow$ ADC ~ 4095)
Expected Output	Channel A ADC: 0, ~ 2048 , ~ 4095 (± 10 counts). Channel B ADC: identical to Channel A within ± 5 counts. No sensor fault flags raised within valid range.
Pass Criterion	Both channels independently read correct ADC values within tolerance; cross-channel ADC values agree within ± 5 counts; no faultADC flag raised during valid range input
Coverage Target	Hardware path coverage of ADC peripheral register access on both channels
Status	NOT TESTED

4.3.2 TC-VER-006 — GPIO Output State and Deployment Command Timing

Table 14: TC-VER-006

Test ID	TC-VER-006
Title	GPIO Output State Readback and Deployment Command Timing
FSRS Reference	FSR-03, FSR-05, FSR-06
ASIL / SIL	ASIL D / SIL 3
Method	Issue a software-commanded deployment signal via debug injection (bypassing threshold check for test purposes). Measure output GPIO pin voltage with oscilloscope on D8 (Airbag1 vote) and D9 (Airbag2 vote). For side-airbag timing (FSR-06): apply step input on POTI-3 and measure time from threshold crossing to D8 HIGH using oscilloscope timestamp.
Input Conditions	(a) Deployment flag = FALSE: D8 and D9 expected LOW (< 0.4 V). (b) Deployment flag = TRUE (debug override): D8 expected HIGH (> 2.9 V). (c) Side-impact stimulus: POTI-3 stepped above ADC 1638 threshold.
Expected Output	(a) Measured voltage < 0.4 V for D8 and D9. (b) Measured voltage > 2.9 V for D8; D9 follows based on babySeat state. (c) D8 assertion within < 8 ms of POTI-3 threshold crossing.
Pass Criterion	Electrical state matches software intent in both conditions (a) and (b); timing from command to output < 1 ms via oscilloscope; side-impact response < 8 ms (FSR-06)

Coverage Target	End-to-end software-to-hardware output path; side-impact interrupt path
Status	NOT TESTED

4.3.3 TC-VER-007 — CAN Frame Transmission Hardware Integration

Table 15: TC-VER-007

Test ID	TC-VER-007
Title	CAN Frame Transmission and Reception at OMS Lab PC — Hardware Integration
FSRS Reference	FSR-13, FSR-14, FSR-15, FSR-16, FSR-16A
ASIL / SIL	ASIL B / SIL 2 (ASIL D end-to-end per FSR-16A)
Method	Trigger crash simulation on both channels (POTI-1 and POTI-3 above threshold). Capture transmitted CAN frames at OMS lab PC using USBCANV2 CAN analyser at 500 kbit/s. Verify frame ID (0x100 or 0x200), data byte layout, CRC field (bytes 6–7), and inter-frame timestamp.
Input Conditions	Crash simulation: POTI-1 → ADC > 1228 (> 30 g); POTI-3 → ADC > 1638 (> 200 kPa); baby-seat switch HIGH (adult). Both channels active and CAN bus connected.
Expected Output	CAN frame ID = 0x100 (ECU1) or 0x200 (ECU2). Byte 0 Bit 0 = 1 (driver deploy); Byte 0 Bit 1 = 1 (pass deploy when no suppression); Byte 0 Bit 3 = 1 (crash detected). CRC bytes 6–7 valid. Frame timestamp within 15 ms of POTI threshold crossing.
Pass Criterion	All specified bit fields correct; CRC verification passes at receiver; parallel readback discrepancy count = 0; transmission timestamp ≤ 15 ms from threshold crossing; heartbeat frame interval = 500 ms ±50 ms over 20 consecutive frames
Notes	*GAP-03: CRC check portion BLOCKED until CRC is implemented. *GAP-04: Parallel readback check BLOCKED until readback channel is implemented. Frame ID, byte layout, and timing sub-tests can proceed.
Status	NOT TESTED (partially BLOCKED pending GAP-03, GAP-04)

4.3.4 TC-VER-008 — Power-On Self-Test Complete Sequence

Table 16a: TC-VER-008

Test ID	TC-VER-008
Title	Power-On Self-Test Complete Sequence Timing and Coverage
FSRS Reference	FSR-17
ASIL / SIL	ASIL C / SIL 2
Method	Power-cycle the Arduino Due system. Measure total POST duration using Serial timestamp output (millis() at POST start vs. millis() at completion). Verify each POST item (CPU, RAM, ADC, clock, CAN, baby-seat, power supply, DWI) produces a PASS log entry on Serial Monitor.
Input Conditions	All sensors within valid range; CAN bus connected and peer ECU responsive; baby-seat switch HIGH; power supply 12 V nominal.

Expected Output	POST completion time < 800 ms. POST items logged: cpuTest PASS, ramTest PASS, adcTest PASS, clock check PASS, CAN init PASS, baby-seat continuity PASS, supply voltage PASS, DWI lamp PASS. No DTC raised. DWI not illuminated. System transitions to ST-03 (MONITORING).
Pass Criterion	POST completes in < 800 ms; all nine checks pass; no spurious DTCs; DWI remains OFF; Serial output shows system enters ST-03; firmware state = SS-02 (Deployment Armed)
Notes	*GAP-05: CAN communication not verified during POST in current firmware (Can0.begin() success not checked). CAN POST sub-test is BLOCKED until GAP-05 is resolved. CPU, RAM, and ADC POST sub-tests can proceed.
Status	NOT TESTED (partially BLOCKED pending GAP-05)

4.3.5 The Static Code & Watchdog Placement Test

This formalizes the verification of your mandatory coding constraints (CC01-CC08) and proves that the watchdog feed isn't hidden behind a conditional statement.

Table 16b : TC-VER-009

Field	Detail
Test ID	TC-VER-009
Title	Static Code Analysis — Coding Constraints and Watchdog Placement
FSRS Reference	FSR-02, FSR-22
ASIL / SIL	ASIL D / SIL 3
Module Under Test	Entire Firmware Codebase (Static Analysis / Manual Review)
Method	Automated static code analysis and manual code review of the complete project Git repository prior to compilation.
Input Conditions	Firmware source code files (.ino, .c, .h).
Expected Output	Zero instances of malloc, new, delay(), or recursive calls. All THRESH_x values defined as const. Physical unit divisor is explicitly 4095.0. watchdog_feed() is unconditionally the first executable statement in loop().
Pass Criterion	100% compliance with Table 8 (Mandatory Coding Constraints CC01-CC08) of SRS-AIRBAG-SW-001 Rev 2.1.
Coverage Target	100% adherence to safety coding guidelines.
Notes	Captures constraint verifications that cannot be proven via dynamic execution.
Status	NOT TESTED

4.3.6 The Physical Hardware AND Gate Verification

This ensures that a rogue software fault outputting "HIGH" on a single board cannot bypass the physical 2oo2 voting mechanism, and verifies the 3.3V domain safety.

Table 16c: TC-VER-010

Field	Detail
Test ID	TC-VER-010
Title	Hardware AND Gate Truth Table and Voltage Domain Verification
FSRS Reference	FSR-03, FSR-08
ASIL / SIL	ASIL D / SIL 3
Module Under Test	DM74LS08N Hardware Logic & Power Distribution
Method	Part A: Disconnect D8 inputs from Arduino Dues. Manually inject HIGH (3.3 V) and LOW (0 V) signals directly into DM74LS08N pins 1 and 2 in all combinations. Observe Airbag1 LED. Part B: Measure voltage at POT1 leg 1 and D11 pull-up with a calibrated multimeter.
Input Conditions	Part A: Four states for pins 1 & 2: (0 V, 0 V), (3.3 V, 0 V), (0 V, 3.3 V), (3.3 V, 3.3 V). Part B: System powered normally.
Expected Output	Part A: LED is OFF and pin 3 is < 0.4 V for the first three states. LED is ON and pin 3 is > 2.7 V ONLY when both inputs are 3.3 V. Part B: POT1 leg 1 = 3.3 V. D11 open state = 3.3 V.
Pass Criterion	Physical AND gate strictly enforces 2oo2 voting truth table independently of software state. No 5 V signals enter 3.3 V-limited Arduino Due domains.
Coverage Target	Hardware logic physical layer; 100% state coverage of the DM74LS08N.
Notes	Verifies resolution of HADS-CR-001 and HADS-OB-002 voltage constraints.
Status	NOT TESTED

4.3.7 The Main Loop Timing Test

While your summary table listed < 1 ms, you needed an explicit test case to prove it under load.

Table 16c: TC-VER-011

Field	Detail
Test ID	TC-VER-011
Title	Main Loop Cycle Time Verification (Deterministic Execution)
FSRS Reference	FSR-02
ASIL / SIL	ASIL D / SIL 3
Module Under Test	loop() cyclic executive timing
Method	Instrument the test firmware by recording micros() at the very start of loop() and at the very end. Calculate the difference and output the maximum observed duration to the Serial Monitor over 10,000 continuous cycles. Apply maximum sensor load during the test.
Input Conditions	System under heavy load: POT1 1, 2, and 3 constantly varying across thresholds, continuous CAN traffic, and baby-seat switch toggling to force all conditional branches to evaluate.
Expected Output	The maximum recorded cycle time does not exceed 1000 microseconds (1 ms) under any tested condition.
Pass Criterion	Worst-case execution time (WCET) ≤ 1 ms, satisfying FSR-02 (Algorithm Cycle Time).
Coverage Target	Timing coverage of main cyclic executive.
Notes	Validates the absence of blocking logic (e.g., delay()) during runtime.
Status	NOT TESTED

4.4 FSR-15 CAN Timeout — Known Deviation

DEVIATION NOTICE: FSR5 FSR-15 requires CAN communication timeout detection within 50 ms. SADS-AIRBAG-ECU-001 Rev 2.0 Section 2.3 implements a 2000 ms heartbeat timeout (faultCOMM = TRUE when millis() - lastRemoteHB > 2000 ms). This exceeds the 50 ms specification by a factor of 40×. This deviation is documented in the SADS RTM (Section 9) as an open item against FSR-15. Until a design change or formal acceptance rationale is approved, TC-VAL-008 shall verify the implemented 2000 ms behaviour and note the FSR-15 non-conformance. An NCR shall be raised against this deviation.

5 Validation Plan (IEC 61508 Phases 7, 8 and 9)

System-level black-box validation cases are executed on the complete integrated test board to prove the system correctly mitigates the hazards identified in LCP03 HARA. Validation cases treat the ECU as a black box — input stimuli applied via potentiometer positions and digital switch states; outputs observed via CAN bus frames and DWI indicator state.

5.1 Boundary Value Validation Test Cases

TC-VAL-001 — All Sensors at Minimum Boundary (No-Deploy)

Table 17: TC-VAL-001

Test ID	TC-VAL-001
Title	All Sensors at Minimum Boundary — No Deployment
Safety Goal	SG-01 — No unintended deployment
Governing Hazard	H1-01, H1-02
ASIL / SIL	ASIL D / SIL 3
Preconditions	POST complete; system in ST-03 (MONITORING) / SS-02; baby-seat switch HIGH (adult mode); both channels operational
Input Stimuli	POTI-1 = 0 V (ADC ~0, 0 g); POTI-2 = 0 V (ADC ~0, 0 °/s); POTI-3 = 0 V (ADC ~0, 0 kPa). Both ECU channels at minimum sensor values.
Expected Serial Output	localCrash=0, remoteCrash=0, systemFault=0, voteDeploy=0, D8=LOW, D9=LOW on both ECUs
Expected CAN Output	Byte 0 = 0x00 (no deployment bits set); Bit 3 = 0 (no crash detected)
Expected DWI State	OFF
Pass Criterion	No deployment CAN frame transmitted during 10-second monitoring period; Serial output shows crash_detected = 0 on both channels; DWI remains OFF; AND gate output confirmed LOW
Status	NOT TESTED

TC-VAL-002 — All Sensors at Maximum Boundary (Full Deploy)

Table 18: TC-VAL-002

Test ID	TC-VAL-002
Title	All Sensors at Maximum Boundary — Full Deployment Command
Safety Goal	SG-02 — Correct deployment during crash
Governing Hazard	H2-01, H2-02
ASIL / SIL	ASIL D / SIL 3
Preconditions	POST complete; system in SS-02; baby-seat switch HIGH (adult mode); both channels operational
Input Stimuli	POTI-1 = 3.3 V (ADC ~4095, 100 g); POTI-2 = 3.3 V (ADC ~4095, 360 °/s); POTI-3 = 3.3 V (ADC ~4095, 500 kPa); both channels simultaneously at maximum.
Expected CAN Output	Byte 0: Bit 0 = 1 (driver deploy); Bit 1 = 1 (pass deploy); Bit 2 = 0 (no suppression); Bit 3 = 1 (crash detected). CRC bytes 6–7 valid (when GAP-03 resolved). Frame received at OMS PC within 40 ms of threshold crossing.

Expected DWI State	OFF (no fault — valid crash event)
Pass Criterion	Deployment CAN frame received within 40 ms; both airbag bits set; Serial confirms voteDeploy=1 on both ECUs; AND gate fires (Airbag1 and Airbag2 LEDs ON if no baby-seat suppression); system latches post-deployment
Status	NOT TESTED

TC-VAL-003 — POTI-1 Stepped Through Deployment Threshold

Table 19: TC-VAL-003

Test ID	TC-VAL-003
Title	POTI-1 Stepped Through Deployment Threshold — Exact Boundary (30 g / ADC 1228)
Safety Goal	SG-01, SG-02
Governing Hazard	H1-01, H2-01
ASIL / SIL	ASIL D / SIL 3
Preconditions	POTI-2 and POTI-3 below threshold; baby-seat switch HIGH; system in SS-02; both channels active
Input Stimuli	POTI-1 stepped from 0 V to 3.3 V in increments. Key test points: ADC 1227 (below threshold); ADC 1228 (at threshold); ADC 1229 (above threshold). Threshold voltage = $(1228/4095) \times 3.3 \text{ V} = 0.990 \text{ V}$.
Expected CAN Output	At $\text{ADC} < 1228$: no deployment (Byte 0 = 0x00). At $\text{ADC} \geq 1228$: deployment command issued on both channels (Byte 0 Bit 0 = 1). Threshold response repeatable across 3 consecutive test runs.
Expected DWI State	OFF throughout (valid operation, no fault)
Pass Criterion	No deployment below ADC 1228; deployment occurs at $\text{ADC} \geq 1228$; response is repeatable; no hysteresis anomalies observed
Status	NOT TESTED

5.2 Critical Safety Use Cases

TC-VAL-004 — Use Case A: Simulated Frontal Crash — Full Deployment

Table 20: TC-VAL-004

Test ID	TC-VAL-004
Title	Use Case A — Simulated Frontal Crash: Driver and Passenger Airbag Deployment
Safety Goal	SG-02
Governing Hazard	H2-01
ASIL / SIL	ASIL D / SIL 3
Preconditions	POST complete; system in SS-02; baby-seat switch HIGH (adult occupant); both Channel A and Channel B operational; CAN bus connected to OMS PC
Input Stimuli	POTI-1 advanced to maximum ADC (~4095, 100 g) — simulates frontal crash pulse > 30 g. POTI-3 advanced to $\text{ADC} > 1638$ (> 200 kPa). Baby-seat switch remains HIGH throughout.

Expected CAN Output	Byte 0 = 0x0B: Bit 0 (driver)=1, Bit 1 (pax)=1, Bit 2 (suppress)=0, Bit 3 (crash)=1. CRC valid. Frame at OMS PC within 40 ms.
Expected DWI State	OFF during deployment (valid crash — not a fault condition). DWI behaviour post-deployment per firmware implementation.
Pass Criterion	Both driver and passenger airbag deployment bits set in CAN frame; deployment within 40 ms; CRC valid (when GAP-03 resolved); Serial confirms voteDeploy=1 on both channels; AND gate fires; Airbag1 LED ON, Airbag2 LED ON
Status	NOT TESTED

TC-VAL-005 — Use Case B: Baby Seat Present — Passenger Airbag Suppressed

Table 21: TC-VAL-005

Test ID	TC-VAL-005
Title	Use Case B — Baby Seat Present: Passenger Airbag Suppressed During Simulated Crash
Safety Goal	SG-03
Governing Hazard	H1-04
ASIL / SIL	ASIL D / SIL 3
Preconditions	POST complete; system in SS-02; baby-seat switch LOW (baby seat detected / D11 = LOW); both channels operational
Input Stimuli	Baby-seat switch maintained LOW. POTI-1 advanced to ADC > 1228 (> 30 g). POTI-3 advanced to ADC > 1638 (> 200 kPa). Crash conditions met on both channels.
Expected CAN Output	Byte 0: Bit 0 = 1 (driver DEPLOY); Bit 1 = 0 (passenger SUPPRESSED); Bit 2 = 1 (suppression active); Bit 3 = 1 (crash detected). CRC valid.
Expected DWI State	OFF. Baby-seat active suppression is normal operation per LCP05 SCD Section 7.4 — it is NOT a fault condition. DWI shall NOT illuminate for active baby-seat suppression.
Pass Criterion	Driver airbag deploys (Airbag1 LED ON); passenger airbag suppressed (Airbag2 LED OFF, Byte 0 Bit 1 = 0, Bit 2 = 1); CAN frame received within 40 ms; DWI remains OFF; SG-03 satisfied
Notes	CORRECTION from draft V1.0: The draft incorrectly stated DWI = ON for active baby-seat suppression. Active suppression (babySeat=TRUE) is the intended fail-safe default — it is not a fault. DWI activates only for fault conditions (systemFault=TRUE). Reference: SADS Section 4.4.1, Deployment Control logic.
Status	NOT TESTED

TC-VAL-006 — Use Case B2: Baby Seat Line Open-Circuit — Suppress-on-Fault

Table 22: TC-VAL-006

Test ID	TC-VAL-006
Title	Use Case B2 — Baby Seat Line Open-Circuit: Suppress-on-Fault Strategy
Safety Goal	SG-03

Governing Hazard	H1-04 (stuck-HIGH fault)
ASIL / SIL	ASIL D / SIL 3
Preconditions	System armed; baby-seat switch initially HIGH (adult mode); CAN bus connected
Input Stimuli	Physically disconnect the baby-seat digital input wire (simulates open-circuit / stuck-HIGH fault per GAP-02). Then advance POTI-1 beyond deployment threshold (ADC > 1228).
Expected Output	After line disconnect: pull-up holds D11 HIGH → babySeat=FALSE in current firmware (GAP-02: line continuity check not implemented, so stuck-HIGH not actively detected). During crash simulation: passenger airbag bit = 0 (suppressed only if babySeat forced TRUE by continuity check — see GAP-02). DWI: ON within 800 ms if continuity check detects fault (GAP-02 required).
Expected DWI State	ON within 800 ms (requires GAP-02 continuity check to be implemented)
Pass Criterion	When GAP-02 is resolved: DWI activates within 800 ms of disconnect; DTC logged for baby-seat line fault; passenger airbag suppressed despite input reading HIGH; driver airbag remains armed. Until GAP-02 resolved: this test is BLOCKED for the suppress-on-fault aspect.
Notes	*GAP-01: Debounce not implemented — affects TC-VAL-006 signal stability sub-test. *GAP-02: Line continuity monitoring not implemented — primary reason this test is partially BLOCKED.
Status	NOT TESTED (BLOCKED pending GAP-01 and GAP-02)

TC-VAL-007 — Use Case C(i): Sensor Out-of-Range — DWI and Safe State

Table 23: TC-VAL-007

Test ID	TC-VAL-007
Title	Use Case C(i) — POTI Sensor Out-of-Range: DWI Activation and Safe State
Safety Goal	SG-06
Governing Hazard	H4-01
ASIL / SIL	ASIL C / SIL 2
Preconditions	System in normal operation (SS-02 / ST-03); baby-seat switch HIGH
Input Stimuli	Force POTI-1 input above 3.3 V (e.g., connect to 5 V supply) or disconnect to create open-circuit (ADC reads 0 or max rail at out-of-range detection boundary). Maintain for > 200 ms.
Expected CAN Output	Byte 4 sensor fault bits set for POTI-1; Byte 0 Bit 6 (DWI active) = 1; DTC for sensor out-of-range stored
Expected DWI State	ON within 800 ms of fault detection (FSR-19)
Pass Criterion	DWI activates within 800 ms; DTC logged; affected sensor channel isolated (SS-04 state); remaining channels continue monitoring; no spurious deployment command issued during fault condition
Status	NOT TESTED

TC-VAL-008 — Use Case C(ii): CAN Timeout — Limp-Home Mode

Table 24: TC-VAL-008

Test ID	TC-VAL-008
Title	Use Case C(ii) — CAN Bus Communication Timeout: Limp-Home / Fault Mode
Safety Goal	SG-06, SG-07
Governing Hazard	H4-01, H4-02
ASIL / SIL	ASIL B / SIL 2
Preconditions	CAN bus active; both ECUs transmitting heartbeat frames at 500 ms interval; OMS PC receiving frames normally
Input Stimuli	Physically disconnect the CAN bus between ECU1 and ECU2 (remove twisted pair / SN65HVD230 connection). Maintain disconnection for > 2100 ms.
Expected Output	After 2000 ms timeout: faultCOMM=TRUE → systemFault=TRUE → voteDeploy=FALSE → D8/D9 LOW. DWI (D10) HIGH. System enters ST-04 (FAULT_ACTIVE) / SS-01 (deployment inhibit).
Expected DWI State	ON within 2000 ms + 1 cycle (approximately 2001 ms from last valid frame)
Pass Criterion	faultCOMM declared ≤ 2050 ms after last valid frame; DWI activates; DTC logged; no unintended deployment or inhibition occurs during timeout period; system recovers to ST-03 when CAN reconnected (TC-09 equivalent)
Notes	FSR-15 DEVIATION: FSRs requires timeout ≤ 50 ms; current implementation uses 2000 ms. This test validates the implemented 2000 ms behaviour. An NCR is raised against this deviation (see Section 1.2 Unresolved Topics).
Status	NOT TESTED (NCR required for FSR-15 deviation)

TC-VAL-009 — Single-Channel Fault: Channel A Forced Fault

Table 25: TC-VAL-009

Test ID	TC-VAL-009
Title	Single-Channel Fault Injection — Channel A Forced Fault, Channel B Continues
Safety Goal	SG-01
Governing Hazard	H1-01
ASIL / SIL	ASIL D / SIL 3
Preconditions	Both channels operational; system in SS-02; CAN bus connected; OMS PC monitoring
Input Stimuli	Inject a software fault flag on Channel A (e.g., temporarily modify cpuTest() to return false — see SADS Section 4.1.1 WB test). Simultaneously advance POTI-1 beyond ADC 1228 (> 30 g) on both ECUs.
Expected Output	Channel A: faultCPU=TRUE → systemFault=TRUE → voteDeploy=FALSE → D8=LOW. Channel A CAN frame Byte 0 Bit 4 (Ch-A fault) = 1. 2oo2 AND gate: both D8 pins must be HIGH for deployment — Channel A D8=LOW prevents AND gate from firing. No deployment output. DWI ON.

Expected DWI State	ON
Pass Criterion	No deployment command issued with single-channel fault active; DWI ON within 800 ms; AND gate remains LOW (Airbag1 LED OFF); Byte 0 Bit 4 = 1 in CAN frame from Channel A; SC-03 (no single-point fault causes deployment) verified; FSR-07 and FSR-08 satisfied
Status	NOT TESTED

TC-VAL-010 — EMC Disturbance Rejection: Single Sensor Spike

Table 26: TC-VAL-010

Test ID	TC-VAL-010
Title	Single-Sensor Spike Rejection — EMC Disturbance Simulation
Safety Goal	SG-05
Governing Hazard	H1-05
ASIL / SIL	ASIL C / SIL 2
Preconditions	System in SS-02; all sensors below threshold; baby-seat switch HIGH; both channels operational
Input Stimuli	Rapidly advance POTI-1 alone above the deployment threshold (ADC > 1228) on ECU1 only. POTI-1 on ECU2 remains below threshold. POTI-2 and POTI-3 remain below threshold on both ECUs. Hold for 500 ms.
Expected Output	ECU1: localCrash=1 (POTI-1 exceeded threshold on ECU1). ECU2: localCrash=0 (POTI-1 below threshold on ECU2). ECU1 remoteCrash=0 (from ECU2 CAN frame). ECU1 voteDeploy=FALSE (remoteCrash=0 fails the AND-gate). D8=LOW on ECU1. AND gate cannot fire — both D8 outputs NOT simultaneously HIGH. DWI: OFF (no system fault — single-sensor above threshold is not a fault condition).
Expected DWI State	OFF. Single-channel crash detection with no peer agreement is not a fault — it is the expected 2oo2 protection mechanism.
Pass Criterion	No deployment output (AND gate remains LOW); Serial on ECU1 shows localCrash=1 but voteDeploy=0; Serial on ECU2 shows localCrash=0; DWI remains OFF; SG-05 and FSR-04 verified; 2oo2 EMC protection demonstrated
Notes	CORRECTION from draft V1.0: The draft incorrectly implied single-sensor above threshold triggers localCrash=FALSE. Per SADS Section 4.3.2, localCrash = TRUE if ANY single sensor exceeds its threshold. However, the 2oo2 architecture prevents deployment because remoteCrash (from the peer ECU) = FALSE, making voteDeploy = FALSE. The safety is provided by the 2oo2 vote, not by sensor fusion requiring multiple simultaneous thresholds within a single channel.
Status	NOT TESTED

6 Timing Requirements Verification Summary

The following timing parameters shall be verified and validated during V&V execution. All measurements shall be performed with calibrated instrumentation (oscilloscope or CAN analyser with timestamp resolution ≤ 0.1 ms).

Table 27: Timing Requirements Verification

Function	Max Allowed	Verification Method	Test Case(s)
Crash detection algorithm cycle	≤ 1 ms	micros() start/end of loop(); 99th percentile over 10,000 iterations	TC-VER-011
Crash detection response (threshold to decision)	≤ 20 ms	POTI threshold crossing vs. localCrash=TRUE in Serial log	TC-VAL-003, TC-VAL-004
Deployment command transmission (threshold to CAN TX)	≤ 40 ms	CAN analyser timestamp vs POTI crossing	TC-VAL-002, TC-VAL-004
Side-impact deployment logic	< 8 ms	Oscilloscope: POTI-3 step input to D8 HIGH edge	TC-VER-006
Pretensioner activation (if implemented)	< 5 ms	Output GPIO timing measurement	TC-VER-006
CAN timeout detection (FSR-15 specified)	≤ 50 ms	Disconnect CAN; measure faultCOMM assertion time	TC-VAL-008 *DEVIATION: impl=2000 ms
Diagnostic cycle interval	≤ 100 ms	Background task Serial timestamp logging	TC-VER-002, TC-VAL-007
POST execution (complete before arming)	< 800 ms	Serial timestamp: POST start vs. ST-03 entry	TC-VER-008
DWI activation after fault detection	≤ 800 ms	DWI GPIO rising edge vs. fault detection timestamp	TC-VAL-006, TC-VAL-007
Baby-seat debounce stability	≥ 10 ms stable	Signal generator + oscilloscope (when GAP-01 resolved)	TC-VAL-006 *GAP-01
Crash notification CAN frame	≤ 15 ms from deployment auth	CAN analyser deployment frame timestamp	TC-VER-007
CAN heartbeat interval	500 ms $\pm$ 50 ms	USBCANv2 inter-frame timestamps; 20 frames	TC-VER-007

7 Test Report Framework

7.1 Test Case Record Template

Every executed test case shall produce a completed Test Case Record using the following standardised template. Test records shall be archived as formal safety evidence artefacts.

Table 28: Test Case Record Template

Test Case ID	TC-VER-xxx / TC-VAL-xxx
Test Case Title	<Short descriptive title>
Date of Execution	YYYY-MM-DD
Executed By (Test Engineer)	<Name / Role — must be independent of development>
Reviewed By (V&V Lead)	<Name / Role>
FSRS Requirement(s)	FSR-xx, FSR-yy
Safety Goal(s)	SG-xx
Governing Hazard(s)	Hx-xx
ASIL / SIL	ASIL x / SIL x
Hardware Configuration	Arduino Due Channel A: FW Rev x.x Channel B: FW Rev x.x AND gate: DM74LS08N
CAN Interface	OMS Lab PC USB-CAN-A interface model: <model> USBCANV2 at 500 kbit/s
POTI-1 Setting (Accelerometer)	<ADC count> → <Equivalent g value> (threshold boundary if applicable)
POTI-2 Setting (Gyroscope)	<ADC count> → <Equivalent °/s value>
POTI-3 Setting (Pressure Sensor)	<ADC count> → <Equivalent kPa value>
Baby-Seat Switch State	HIGH (adult) / LOW (baby seat) / Disconnected (fault)
Expected CAN Frame (Byte 0–Byte 7)	<Hex values, e.g., 0x0B 0x01 0xAB 0xCD 0x00 0x00 0xCR 0xCC>
Expected DWI State	ON / OFF — <Rationale>
Expected Timing	<Max allowed time, e.g., Deployment within 40 ms>
Actual CAN Frame Received	<Captured hex values from OMS PC USBCANV2 log>
CRC Verification Status	PASS / FAIL — <Expected CRC vs received CRC> (*GAP-03 if not yet implemented)
Parallel Readback Result	PASS / FAIL — <Discrepancy count> (*GAP-04 if not yet implemented)
Actual DWI State	ON / OFF — <Observed>
Actual Timing Measured	<ms measured from threshold to CAN frame or GPIO edge>
Serial Monitor Snapshot	<Key variables: localCrash, remoteCrash, systemFault, voteDeploy, babySeat, A0, A1, A2>
Deviation / Anomaly Notes	<Any unexpected behaviour, non-conformances>
PASS / FAIL Verdict	✓ PASS ✗ FAIL ⚠ BLOCKED

Non-Conformance Reference	NCR-xxx (if FAIL) / N/A
Signature (Test Engineer)	_____ Date: _____
Signature (V&V Lead)	_____ Date: _____

7.2 Serial Monitor Evidence Collection

For each test case, the Serial Monitor output shall be captured and archived. The following variable set shall be recorded as minimum evidence that deployment thresholds were reached or not reached (from SADS Section 8.5):

Table 29: Required Serial Monitor Variables per Test Case

Variable	Meaning / Evidence Value
ECU=1 or ECU=2	Identifies which channel produced this Serial output
A0=<value>	Raw ADC count for POTI-1 (accelerometer); must be 0–4095 range
A1=<value>	Raw ADC count for POTI-2 (gyroscope)
A2=<value>	Raw ADC count for POTI-3 (pressure sensor)
Baby=<0/1>	Baby-seat switch state: 1=LOW (baby seat), 0=HIGH (adult)
Local Crash=<0/1>	localCrash flag: 1 if any sensor exceeded threshold on this channel
Remote Crash=<0/1>	remoteCrash flag: peer ECU crash detection state from CAN
Comm Fault=<0/1>	faultCOMM flag: 1 if peer CAN timeout exceeded 2000 ms
System Fault=<0/1>	systemFault aggregate flag: OR of faultCPU, faultRAM, faultADC, faultCOMM
Vote Deploy=<0/1>	voteDeploy: final deployment authorization output

7.3 Test Summary Report Structure

Upon completion of all test cases, a Test Summary Report shall be produced consolidating all individual test records. The report shall include:

- Document header: document number, revision, date, system, applicable standards, author
- Test scope summary: total number of test cases planned, executed, passed, failed, blocked
- ASIL D / SIL 3 requirements coverage: table confirming 100% FSR coverage with PASS status
- Safety goal validation summary: table confirming each SG validated by at least one PASS test
- Traceability matrix: final completed RTM with all test statuses populated
- Compliance gap closure: confirmation that GAP-01 to GAP-07 are resolved and re-tested
- Non-conformance summary: list of all NCRs raised (including FSR-15 CAN timeout deviation), status, and resolution
- Residual risks: any identified residual risks not fully mitigated by V&V results
- Approval signatures: Functional Safety Manager, V&V Lead, Project Supervisor

7.4 Non-Conformance Management

Any test case resulting in FAIL or BLOCKED shall trigger the following process:

- Raise a Non-Conformance Report (NCR) with unique identifier (NCR-xxx)
- Classify NCR severity: Critical (ASIL D / SIL 3 requirement violated), Major (ASIL B-C / SIL 1-2), or Minor

-
- Assign corrective action owner and target resolution date
 - Re-execute the affected test case after corrective action implementation
 - V&V Lead reviews and signs off NCR closure; update RTM status from FAIL to PASS

8 Requirements Traceability Matrix (RTM)

The following matrix provides full bidirectional traceability from HARA hazards through safety goals, FSRs requirements, to V&V test cases. All statuses are set to NOT TESTED at plan issue. All ASIL D / SIL 3 requirements shall achieve PASS status before system release. Entries marked *GAP-xx are dependent on firmware compliance gap resolution.

Table 30: Requirements Traceability Matrix

Hazard ID	Safety Goal	ASIL/SIL	FSR ID	FSR Short Title	SC Mechanism	Code Module(s)	Test Case(s)	Coverage Type	Status
H2-01	SG-02	ASIL D / SIL 3	FSR -01	Crash Detection Timing	Cyclic executive ≤ 1 ms	MOD-02, MOD-03	TC-VER-002, TC-VAL-003	BB + Timing	NOT TESTED
H2-04	SG-02	ASIL D / SIL 3	FSR -02	Algorithm Cycle Time	≤ 1 ms loop discipline	All modules	TC-VER-002, TC-VER-011	Timing	NOT TESTED
H1-01	SG-01	ASIL D / SIL 3	FSR -03	Deployment Auth (6-cond)	2oo2 AND-gate vote logic	MOD-04	TC-VER-002, TC-VER-010, TC-VAL-009	BB + WB MC/DC	NOT TESTED
H1-01, H1-02	SG-01	ASIL D / SIL 3	FSR -04	Prevention Unintended Deploy	2oo2 + inhibit logic	MOD-04	TC-VAL-001, TC-VAL-010	BB	NOT TESTED
H2-01	SG-02	ASIL D / SIL 3	FSR -05	Deployment Command Timing	≤ 40 ms from crash detect	MOD-04, MOD-05	TC-VAL-002, TC-VAL-004	Timing	NOT TESTED
H2-02	SG-02	ASIL D / SIL 3	FSR -06	Side-Airbag Timing	< 8 ms interrupt path	MOD-03, MOD-04	TC-VER-006	Timing	NOT TESTED
H1-01	SG-01	ASIL D / SIL 3	FSR -07	Dual-Channel Redundancy	2oo2 architecture	MOD-02–MOD-05	TC-VAL-009	Integration BB	NOT TESTED
H1-01	SG-01	ASIL D / SIL 3	FSR -08	Channel Agreement Verification	Cross-channel CAN vote	MOD-04, MOD-05	TC-VER-010, TC-VAL-009	BB	NOT TESTED
H1-04	SG-03	ASIL D / SIL 3	FSR -09	Passenger Airbag Suppression	Baby-seat suppress logic	MOD-04	TC-VAL-005	BB	NOT TESTED
H1-04	SG-03	ASIL D / SIL 3	FSR -10	Suppression-Fault Strategy	Default suppression on fault	MOD-04	TC-VAL-006	BB Fault Inject	NOT TESTED
H1-04	SG-03	ASIL D / SIL 3	FSR -11	Digital Input Debounce	10 ms stable filter	MOD-02	TC-VAL-006 *GAP-01	BB	NOT TESTED
H1-04	SG-03	ASIL D / SIL 3	FSR -12	Baby Seat Line Monitoring	Continuity check	MOD-06	TC-VAL-006 *GAP-02	BB Fault Inject	NOT TESTED
H4-02	SG-07	ASIL B / SIL 2	FSR -13	CRC-Protected CAN	16-bit CRC over bytes 0–5	MOD-05	TC-VER-004, TC-	Unit + Integration	NOT TESTED

							VER-007 *GAP-03		
H4-02	SG-07	ASIL B / SIL 2	FSR -14	CAN Bitstream Readback	Parallel readback channel	MOD-05	TC-VER-004, TC-VER-007 *GAP-04	Integration	NOT TESTED
H4-02	SG-07	ASIL B / SIL 2	FSR -15	CAN Timeout Detection (≤50 ms)	CAN frame timeout	MOD-05, MOD-06	TC-VAL-008 *NOTE: impl=2000 ms	BB	NOT TESTED
H4-02	SG-07	ASIL B / SIL 2	FSR -16	Crash Notification Timing	CAN TX within 15 ms	MOD-05	TC-VER-007	Timing	NOT TESTED
H1-01, H2-01, H4-02	SG-01/SG-07	ASIL D / SIL 3	FSR -16A	ASIL D CAN End-to-End Integrity	CRC + readback + vote + timeout	MOD-04, MOD-05	TC-VER-004, TC-VER-007, TC-VAL-009	Integration	NOT TESTED
H4-01	SG-06	ASIL C / SIL 2	FSR -17	Power-On Self-Test	POST 800 ms sequence	MOD-01	TC-VER-008 *GAP-05	Integration	NOT TESTED
H4-01	SG-06	ASIL C / SIL 2	FSR -18	Continuous Diagnostics ≤100 ms	Background diagnostic cycle	MOD-06	TC-VAL-007, TC-VAL-008	BB	NOT TESTED
H4-01	SG-06	ASIL A / SIL 1	FSR -19	DWI Activation ≤800 ms	Fault → D10 HIGH ≤800 ms	MOD-04, MOD-06	TC-VAL-006, TC-VAL-007	Timing	NOT TESTED
H1-01	SG-01	ASIL D / SIL 3	FSR -20	Safe-State Transition	systemFault → voteDeploy=FALSE	MOD-04, MOD-06	TC-VAL-007, TC-VAL-009	BB Fault Inject	NOT TESTED
H4-01	SG-06	ASIL A / SIL 1	FSR -21	Fault Logging (DTCs)	DTC to EEPROM/flash	MOD-06	TC-VAL-006, TC-VAL-007, TC-VAL-008 *GAP-07	BB	NOT TESTED
H1-01, H2-01, H4-01	SG-01/SG-06	ASIL D / SIL 3	FSR -22	Memory Integrity Protection	CRC32 RAM + WDT + stack guard	MOD-01, MOD-06	TC-VER-003 GAP-06, TC-VER-009	Unit WB	NOT TESTED
N/A	N/A	ASIL D / SIL 3	Constraints	Code Safety Limits (CC01-CC08)	Static Code Analysis	All modules	TC-VER-009	Static Analysis	NOT TESTED

**GAP notation: Test cases marked *GAP-xx cannot achieve PASS status until the corresponding firmware compliance gap (listed in Section 2.4 and SADS Rev 2.0 Section 7) is resolved and verified. The FSR-15 deviation (CAN timeout 2000 ms vs. 50 ms specified) requires a formal NCR and acceptance rationale.*

9 References and Normative Standards

Table 31: References

Reference	Description
ISO 26262:2018 Parts 1–12	Road Vehicles — Functional Safety
IEC 61508:2010 Parts 1–7	Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems
LCP01-AIRBAG-ECU-001 v1.10	Concept Phase Document — item definition, system boundary, team roles
LCP03-AIRBAG-ECU-001 v2.00	Hazard and Risk Analysis (HARA) — hazards H1-01 to H4-02, safety goals SG-01 to SG-07
LCP04-AIRBAG-ECU-001 v1.00	Functional Safety Requirements Specification (FSRS) — FSR-01 to FSR-22
LCP05-AIRBAG-ECU-001 v1.00	Safety Concept Document (SCD) — safety mechanisms, safe states SS-01 to SS-05
SADS-AIRBAG-ECU-001 Rev 2.0	Software Architecture and Design Specification — module decomposition, test cases TC-01 to TC-18, GAP-01 to GAP-07
ISO 21448:2022	Safety of the Intended Functionality (SOTIF)
ISO 14229-1:2020	Unified Diagnostic Services (UDS)
ISO 11898	Road Vehicles — Controller Area Network (CAN)
SAE J211	Instrumentation for Impact Tests
FMVSS No. 208	Occupant Crash Protection — U.S. Department of Transportation, NHTSA

10 List of Dependent Documents

Table 32 lists all documents referenced by or related to this V&V Plan.

Table 32: Dependent Documents

No.	Document Name	Version	Date	File Name
1	Life Cycle Phase 1 — Concept Document	1.10	2026-05-03	LCP01-AIRBAG-ECU-001_Concept_v110.docx
2	Life Cycle Phase 2 — Task of the Process or Product	1.10	2026-05-03	LCP02-AIRBAG-ECU-001_Concept_v_110.docx
3	Life Cycle Phase 3 — Hazard and Risk Analysis (HARA)	2.00	2026-05-05	LCP03-AIRBAG-ECU-001_HARA_v110.docx
4	Life Cycle Phase 4 — Functional Safety Requirements Specification	1.00	2026-05-11	LCP04-AIRBAG-ECU-001_FSRs_v100.docx
5	Life Cycle Phase 5 — Safety Concept Document	1.00	2026-05-12	LCP05-AIRBAG-ECU-001_SCD_v100.docx
6	Software Architecture and Design Specification	Rev 2.0	2026-06-12	SADS-AIRBAG-ECU-001_Rev2_0.docx
7	ISO 26262:2018 — Road Vehicles — Functional Safety	2018	—	External standard
8	IEC 61508:2010 — Functional Safety of E/E/PE Systems	2010	—	External standard

11 Figure and Table Listing

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- Table 12: TC-VER-004 — CAN Frame and CRC
- Table 13: TC-VER-005 — Sensor ADC Integration
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- Table 15: TC-VER-007 — CAN Transmission Integration
- Table 16a: TC-VER-008 — Power-On Self-Test
- Table 16b: TC-VER-009 — Static Code Analysis
- Table 16c: TC-VER-010 — Hardware AND Gate Truth Table
- Table 16d: TC-VER-011 — Main Loop Cycle Time
- Table 17: TC-VAL-001 — Minimum Boundary No-Deploy
- Table 18: TC-VAL-002 — Maximum Boundary Full Deploy

Table 19: TC-VAL-003 — Threshold Boundary Step Test
Table 20: TC-VAL-004 — Use Case A Frontal Crash
Table 21: TC-VAL-005 — Use Case B Baby Seat Suppression
Table 22: TC-VAL-006 — Use Case B2 Suppress-on-Fault
Table 23: TC-VAL-007 — Use Case C(i) Sensor Out-of-Range
Table 24: TC-VAL-008 — Use Case C(ii) CAN Timeout
Table 25: TC-VAL-009 — Single-Channel Fault Injection
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Table 27: Timing Requirements Verification
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Table 33: Version History

Document Name / File	Date	Change Log
VVP-AIRBAG-ECU-001.docx	2026-05-15	Version 1.0 — Initial draft release. Covered verification test cases TC-VER-001 to TC-VER-008 and validation cases TC-VAL-001 to TC-VAL-010. RTM for FSR-01 to FSR-22. Test report template included.
VVP-AIRBAG-ECU-001_v110.docx	2026-06-12	Version 1.10 — Major revision. Corrections: (1) ADC range corrected from 10-bit (0–1023) to 12-bit (0–4095) throughout, matching Arduino Due hardware configuration in SADS Rev 2.0. (2) TC-VAL-005 DWI state corrected: active baby-seat suppression is normal operation, not a fault — DWI shall be OFF. (3) TC-VAL-010 expected output corrected: single-sensor above threshold triggers localCrash=TRUE on that channel; deployment prevented by 2oo2 vote (remoteCrash=FALSE on peer), not by multi-sensor fusion within one channel. (4) All RTM statuses reset to NOT TESTED (draft incorrectly showed PASS for all entries before any testing). (5) Section 2.4 added: seven firmware compliance gaps GAP-01 to GAP-07 from SADS Rev 2.0 documented and cross-referenced throughout. (6) FSR-15 CAN timeout deviation (50 ms spec vs 2000 ms implementation) formally identified and notated in RTM and TC-VAL-008. (7) Document metadata updated to include SADS Rev 2.0 as input reference. (8) Document style aligned with LCP01 project template.

13 Approval and Sign-Off

Table 34: Approval Signatures

Role	Name	Signature	Date
Functional Safety Manager	Kruthika S. Gowda	_____	_____

Testing Engineer V&V Lead	Shivbaraje Bhosale	_____	_____
Software Safety Engineer	Mariana Chavez Avila	_____	_____
Hardware Safety Engineer	Atharva Avsare / Om Naik	_____	_____
Quality Manager	Thirthashri G. Ganesh	_____	_____
Project Supervisor		_____	_____